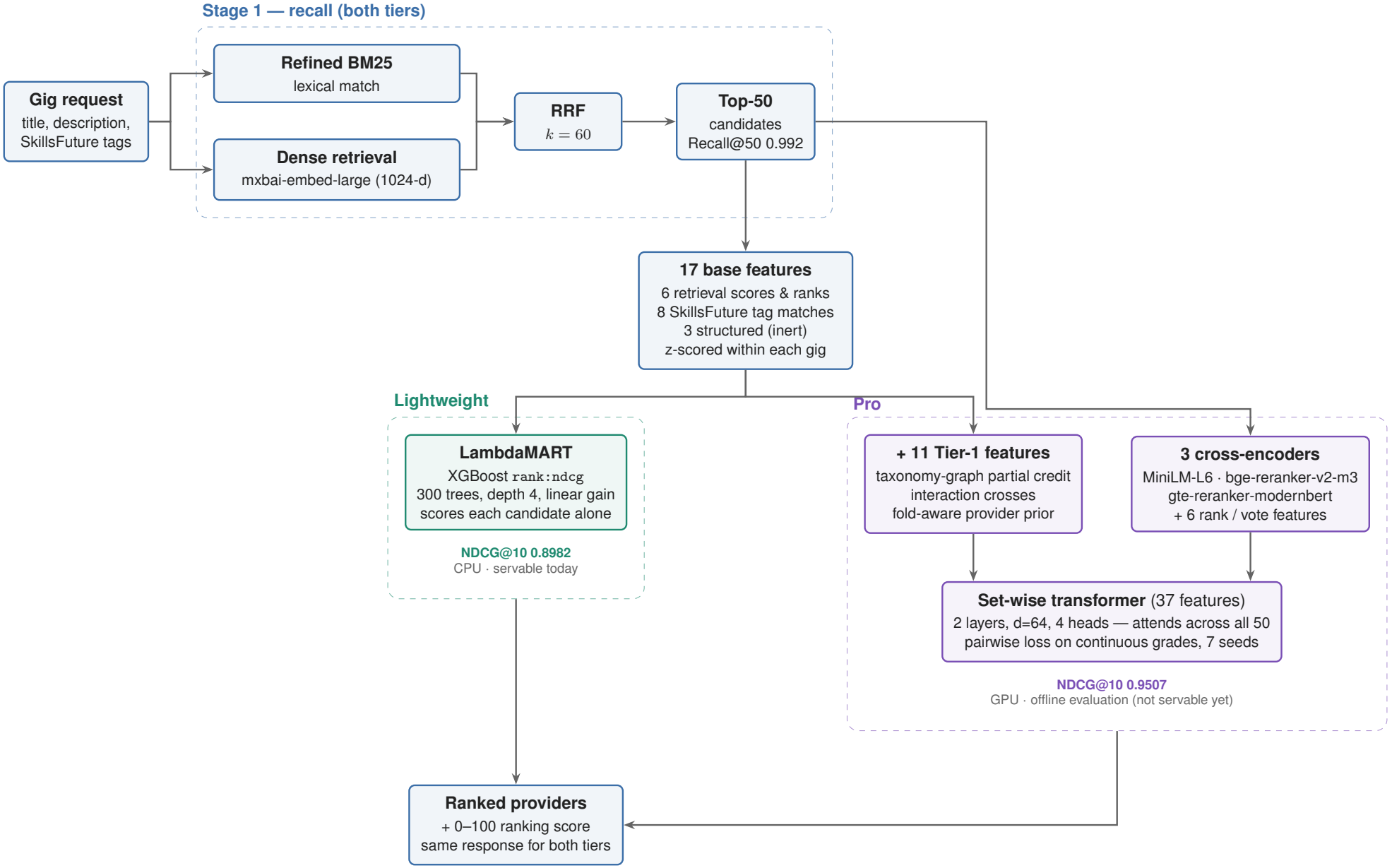

Stage 1 — recall (both tiers)
Gig request
title, description, SkillsFuture tags
Refined BM25
lexical match
Dense retrieval
mxbai-embed-large (1024-d)
RRF
$k = 60$
Top-50
candidates
Recall@50 0.992
17 base features
6 retrieval scores & ranks
8 SkillsFuture tag matches
3 structured (inert)
z-scored within each gig
Lightweight
LambdaMART
XGBoost rank:ndcg
300 trees, depth 4, linear gain
scores each candidate alone
NDCG@10 0.8982
CPU · servable today
Pro
+ 11 Tier-1 features
taxonomy-graph partial credit
interaction crosses
fold-aware provider prior
3 cross-encoders
MiniLM-L6 · bge-reranker-v2-m3
gte-reranker-modernbert
+ 6 rank / vote features
Set-wise transformer (37 features)
2 layers, d=64, 4 heads — attends across all 50
pairwise loss on continuous grades, 7 seeds
NDCG@10 0.9507
GPU · offline evaluation (not servable yet)
Ranked providers
+ 0–100 ranking score
same response for both tiers